

Date: _____

Week 8

October 12, 2026 • 5:00 – 7:30 PM • Kai Santos

NOTES

MECH 191 — Metallurgy · Week 8 Notes · fill in blanks & key terms as you go

01 Surface & Near-Surface Methods

01

02 Volumetric Methods

02

03 Method Selection & Qualification

03

NOTES

EXAMET-5-R-600-HD • 50-500×

Specimens to Check Out

- Specimen A — 1018 steel
- Specimen L — 360 leaded brass
- Specimen P — Ti-6Al-4V titanium

How to check out the kit:

- 1 See instructor after class today
- 2 Sign out the specimen tray
- 3 Use the EXAMET-5 at your convenience
- 4 Return kit before leaving class
- 5 Submit completed module with your portfolio

Portfolio Handout — Module 7

- SLIDE 3** *Metallography Lab — Module 7 EXAMET-5-R-600-HD · 50–500×*

MECH 191 — Course & Unit Roadmap

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Materials Fundamentals	Steel & Alloys	Testing & Quality	Manufacturing	Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 ← here Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

SLIDE 4 MECH 191 — Course & Unit Roadmap

NOTES

Topics

- Visual inspection (VT) — the first step in every inspection sequence; capabilities and limits
- Dye penetrant testing (PT) — capillary action, color contrast vs. fluorescent, dwell time, and surface limitations
- Magnetic particle testing (MT) — ferromagnetic materials only; yoke vs. prod; field orientation and 2-direction rule
- Ultrasonic testing (UT) — sound waves, couplant, straight beam vs. angle beam, A-scan, phased array
- Radiographic testing (RT) — X-ray vs. gamma sources, what volumetric defects look like, planar defect limitation
- Method selection — matching defect type, material, and geometry to the right method
- Written procedures and ASNT SNT-TC-1A qualification levels (I, II, III)

Reference Material

Moniz Ch. 29
Nondestructive Examination

Kalpakjian Ch. 4
Surfaces, Inspection, & Product Quality

Reference Pages

NDT Methods — Physics, Techniques & Limitations

Defect Recognition

Unit 3 Quiz — Weeks 7-9

Scheduled during Week 9

NOTES

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- 1 What is the difference between destructive and non-destructive testing, and why does it matter in practice?
- 2 What are the five main NDT methods and what types of defects can each detect?
- 3 How do you choose the right NDT method for a given situation — what factors must you consider?
- 4 What role do written procedures and technician qualification play in NDT — and why are they non-negotiable?

SLIDE 6 Core Questions

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What VT Can and Cannot Find

Naked eye + adequate lighting, mirrors, and magnifying lenses — the simplest and most universal method. Minimum illuminance is specified in the procedure.

Borescopes, videoscopes, and drones reach where direct access is impossible — internal pipe surfaces, turbine blades, vessel interiors.

Surface cracks, corrosion, mechanical damage, weld surface defects (undercut, overlap, spatter, surface porosity), incorrect geometry, misalignment, coating condition.

Anything subsurface — a visually clean surface gives zero information about what lies beneath. VT is always Step 1; it is never the only step for critical components.

- Check weld profile against the weld symbol on the drawing — correct size, length, and location
- Undercut: a groove at the weld toe that reduces effective throat thickness — measured with a fillet gauge
- Overlap: weld metal rolled over base metal without fusion — visible as a rolled edge; planar defect, no acceptable amount
- Excessive spatter: indicates incorrect parameters; may also indicate moisture in consumables
- Surface porosity: visible pores in the weld face — investigate shielding gas coverage and base metal cleanliness
- Cracks: linear indications at the weld surface, HAZ, or crater — any crack is cause for rejection pending investigation
- Key requirement: lighting ≥ 500 lux (or per procedure) — poor lighting is the most common cause of missed surface defects

SLIDE 7 *Visual Inspection (VT) — The Foundation of Every Inspection*

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Process Steps

- 1 Pre-cleaning:**
Remove all paint, scale, oil, and coatings — any contamination can seal the defect opening and prevent penetrant entry
- 2 Penetrant application:**
Apply to the cleaned surface; low-viscosity liquid is drawn into defects by capillary action
- 3 Dwell time:**
Hold time specified in the procedure — too short misses fine defects; too long may dry the penetrant in the defect before developer is applied
- 4 Excess removal:**
Remove surface penetrant without flushing the defect — use a damp cloth (solvent removable) or water rinse (water-washable)
- 5 Developer application:**
Draws trapped penetrant back out of defects — creates a visible bleed-out indication against a white background
- 6 Inspection + post-clean:**
Inspect per procedure; document indications; clean the surface to remove developer and residual penetrant

Color contrast (visible):

Fluorescent:

Limitations

- Only detects surface-breaking defects — subsurface defects are invisible to PT
- Surface must be clean and free of coatings that seal the defect opening
- Porous materials (as-cast surfaces, thermal spray coatings) cannot be inspected — the whole surface bleeds back
- Works on any non-porous material — metals, ceramics, plastics (unlike MT which needs ferromagnetic material)

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Principle & Magnetization Methods

The component is magnetised; at a defect, the magnetic field is forced out of the material creating flux leakage; iron particles (dry or wet suspension) are attracted to the leakage and form a visible indication.

Electromagnetic yoke placed on the surface induces a field between its poles — portable, widely used for weld inspection; detects defects perpendicular to the field.

Electrical contacts pressed onto the surface pass current through the material — creates a circular field; detects defects running between the prods; risk of arc burns at contact points.

Used for magnetising whole components longitudinally or circularly — common in shop environments for valves, fittings, and bar stock.

Carbon steels, low-alloy steels, martensitic and ferritic stainless steels. Cannot be used on austenitic stainless, aluminum, copper, or titanium.

Detects surface and near-surface defects up to approximately 2-3 mm below the surface — superior to PT for near-surface detection in magnetic materials.

A defect must be at 45–90° to the field to produce a detectable indication. Inspections must be performed in at least two perpendicular directions to ensure all orientations are covered.

Wet fluorescent MT: iron particles in liquid carrier viewed under UV light — the most sensitive MT technique; widely used for critical weld inspection in pressure vessel and pipeline fabrication.

Required after inspection on components where residual magnetism would interfere with service — near instruments, or where particles could contaminate a process.

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Principle & Techniques

Transducer angled to direct sound at 45°, 60°, or 70° into the material — used for weld inspection where the sound must be directed toward fusion faces and the weld root.

Limitations:
Rough surfaces reduce coupling efficiency. Coarse-grained materials (some austenitic stainless, castings) scatter sound and make interpretation difficult. Defects parallel to the beam may not reflect sufficient energy.

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Method Selection — Matching the Method to the Problem

Factor	VT	PT	MT	UT	RT
Defect location	Surface only	Surface-breaking only	Surface & near-surface	Surface & deep internal	Surface & deep internal
Material type	Any	Non-porous any material	Ferromagnetic only	Most metals & composites	Most solid materials
Detects planar cracks well?	Depends (surface)	Yes (if surface-breaking)	Yes (near-surface)	Yes (preferred)	Poor (if parallel to beam)
Access required	One side (direct view)	One side	One side	One side	Both sides (source + film)
Produces permanent image record?	No	No	No	A-scan / S-scan digital	Yes (film or digital)
ASNT SNT-TC-1A — Level I: perform under supervision • Level II: perform, interpret & accept/reject • Level III: write procedures & certify					

SLIDE 12 *Method Selection — Matching the Method to the Problem*

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Core Questions — Answers

What is the difference between destructive and non-destructive testing, and why does it matter?

Destructive testing (tensile, hardness, Charpy) consumes or damages the specimen — accurate but the tested piece cannot go into service. NDT inspects the actual component without altering it, enabling 100% inspection and in-service monitoring of critical components.

What are the five main NDT methods and what does each detect?

VT: visible surface defects. PT: surface-breaking defects in any non-porous material. MT: surface and near-surface defects in ferromagnetic materials. UT: surface and deep internal defects via sound waves. RT: volumetric defects (porosity, inclusions) via radiation imaging.

How do you choose the right NDT method?

Consider: (1) defect type and expected location (surface, near-surface, volumetric); (2) material (ferromagnetic or not, porous or not); (3) geometry and access available; (4) required sensitivity; (5) applicable code or standard specifying the method.

What role do written procedures and technician qualification play — and why are they non-negotiable?

Every method must be performed per a qualified written procedure specifying equipment, materials, technique, and acceptance criteria. Technicians must hold ASNT Level I, II, or III certification for the method. An uncertified person cannot accept or reject an indication, and deviating from the procedure invalidates the inspection regardless of the result.

SLIDE 13 Core Questions — Answers

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2

3

SLIDE 14 *Key Takeaways*

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Week 8 Resources

Nondestructive Examination — primary reading

Surfaces, Inspection, and Product Quality Assurance

Physics, techniques, and limitations for all 5 methods

Visual identification of common weld and material defects

October 19, 2026

- What quality standards are and why they exist — ASTM, AWS, ASME, ISO, and API
- Common weld defects and how to recognize them — porosity, LOF, cracks, undercut
- Indication vs. discontinuity vs. defect vs. reject — precise language that matters
- What good inspection documentation looks like and why it is a legal document

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